

The file "data.csv" contains the dataset.

Here, I report the codebook of the dataset for the variable names that are harder to interpret:

cluster: the cluster of the subject based on the median MRT score
(cluster 0 = MRT > 13; cluster 1 = MRT <= 13)
task: whether the subject summed the pins in the faces of the cube with even or odd number of pins
order: 1 if the subject did the test at the beginning of the exp; 2= at the end of the exp
conf: the configuration of pins locations used in the specific trial
subject_orientation: orientation of the subject in absolute coordinates while doing the test (e.g. 'north' if facing North)
pin_facet_1: number of pins in facet 1
.
.
pin_facet_6: number of pins in facet 6
cond_facet_1: kind of face based on pin number (even or odd)
.
.
cond_facet_6: kind of face based on pin number (even or odd)
correct: correct sum of pins for that trial
response: actual response of the subject
accuracy: 1 = correct; 0 = wrong
n_tocchi_trial = number (unfiltered) of active cells of the cube in the whole trial
n_tocchi_facet_1 = number (unfiltered) of active cells in face #1 of the cube in the whole trial
.
.
n_tocchi_facet_6 = number (unfiltered) of active cells in face #6 of the cube in the whole trial
filt_n_tocchi_trial = number (filtered, i.e., explorative touches only) of active cells of the cube in the whole trial
filt_n_tocchi_facet_1 = number (filtered, i.e., explorative touches only) of active cells in face#1 of the cube in the whole trial
.
.
filt_n_tocchi_facet_6 = number (filtered, i.e., explorative touches only) of active cells in face#6 of the cube in the whole trial
touch_density_trial = unfiltered touch frequency (active cells/second) in the whole trial
touch_density_facet_1 = unfiltered touch frequency (active cells/second) in face#1 in the whole trial
.
.
touch_density_facet_6 = unfiltered touch frequency (active cells/second) in face#6 in the whole trial
filt_touch_density_trial = filtered (i.e. explorative touches only) touch frequency (active cells/second) in the whole trial
filt_touch_density_facet_1 = filtered touch frequency (active cells/second) in face#1 in the whole trial
.
.
filt_touch_density_facet_6 = filtered touch frequency (active cells/second) in face#6 in the whole trial
duration_trial = duration of exploration in s

```

dur_facet_1 = duration of exploration in face#1 in the whole trial
.
.
dur_facet_6 = duration of exploration in face#6 in the whole trial
filt_dur_facet_1 = filtered (explorative touches only) duration of
exploration in face#1 in the whole trial
.
.
filt_dur_facet_6 = filtered (explorative touches only) duration of
exploration in face#6 in the whole trial
rot_abs_total = amount of rotation in deg in the whole trial
raw_mean_rot_velocity = unfiltered rotation velocity (deg/s)
filtered_mean_rot_velocity = filtered (removed rotations < 1deg/s)
rotation velocity (deg/s)
facce_esplorate_0.8s = sequence of explored faces
orient_face_deg = faces orientation in spherical coordinates at the
beginning of exploration of the faces reported in 'facce_esplorate_0.8s'.
All cube faces are considered (i.e. list of 6 couples of coordinates for
each initial timepoint of exploration of a face)
orient_face_world_coord = same as above but translated to world labels
(e.g. 'north', 'up', 'west', etc.)
orient_face_local = same as above but translated to labels relative to
the participants (e.g. 'left', 'down', 'rear'...)
timepoints_facce_esplorate_0.8s = timepoints in s of the beginning of
exploration of a face (see 'facce_esplorate_0.8s')
mean_delta_timepoints = mean time to move from one explored face to the
next (mean of 'timepoints_facce_esplorate_0.8s')
std_delta_timepoints = standard deviation of 'mean_delta_timepoints'
transition_local = position of explored face (see 'facce_esplorate_0.8s')
relative to the participants
trans_matrices = matrices expressing the probability (prop) of moving
from one location to another (e.g. from 'front' to 'left')
ritorni_0.8s = number of returns to already explored faces expressed as
array (e.g. [0 0 0 2 0 0] = face#4 explored twice)
ritorni = total number of returns to already explored faces for that
trial (sum of the array coded in 'ritorni_0.8s')
diff_trans = number of different transitions (i.e.  $n^{\circ}$  of non-zero
elements of the transition matrix)
MRT = visual mental rotation test score

```